Carlos Ojeda-de-Silva

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Education

Harvey Mudd College August 2023 – May 2027
Bachelor of Science in Engineering (GPA: 3.5 / 4.0) *Claremont, CA*

Skills

Languages: Java, Python, C/C++, System Verilog, MATLAB
Developer Tools: SolidWorks, Altium Designer, VS Code, Git, Lattice Radiant, Questa, Google Colab, Jupyter Notebook, Arduino IDE, SEGGER
Additional Skills: CAD, PCB Design, Machining, Single Board Computers, Microcontrollers, FPGA, 3D Printing, Laser Cutting, Finite Element Analysis, Gantt Charting, Project Road-mapping, Technical Writing

Experience

Technical Lead, Electrical and Computer Engineer August 2026 – Present
Blackrock Neurotech *Claremont, CA*
- Engineer a custom ESP32 PCB for BCI data relay, documenting design alternatives and task roadmaps.
- Evaluate power metrics to guarantee continuous multi-hour runtime for neuro-telemetry systems.

Lead Biomedical Engineer January 2025 – Present
CuTE Lab at Harvey Mudd College *Claremont, CA*
- Engineered a PDMS microfluidic device and a custom 4-channel pump to precisely control fluidic resistance, flow rate and mixing ratios.
- Retrofitted a 3D printer for PEG-based bioinks and built a custom toolpath slicer to minimize stiffness gradients.
- Led manufacturing workshops, authored technical memos and process routers for the team.

Subteam Lead, Electrical and Computer Engineer August 2023 – Present
HMC Liquid Rocket Engine *Claremont, CA*
- Architect electrical systems for telemetry and propellant feed control on a liquid rocket test stand.
- Lead the electrical and software teams, delegating tasks and managing project execution timelines.
- Manufacture a liquid rocket engine designed to produce over 1,000 lbf of thrust.

Mechanical Engineer August 2025 – December 2025
Northstar Radioisotopes *Claremont, CA*
- Engineered specialized robotic maintenance tools to drastically reduce worker radiation exposure.
- Designed and tested a dual-function tool to disconnect and cut steel pipes with minimal burring.

Mechanical Engineer November 2023 – December 2024
Sokil Imaging *Claremont, CA*
- Engineered and validated lightweight magnetometer mounts, significantly reducing mechanical oscillation and noise.
- Patented a drone sensor mount currently deployed in Ukraine for magnetometry-based de-mining.

Aerospace Engineer August 2021 – June 2023
Alamo Heights High School Rocketry Program *San Antonio, TX*
- Manufactured custom components to safely recover a 140-lb rocket from an altitude of 50,000 ft.
- Presented flight simulation data to NASA engineers and successfully launched a supersonic vehicle to 18,994 ft.

Projects

Compressed Video Streaming | *Jupyter Notebook, ESP32, LoRa* May 2024 – Present
- Reduce LoRa bandwidth requirements using neural network autoencoder compression.
- Evaluate PCA and Frequency Domain Compression algorithms to maximize data bandwidth.
- Architect a pipeline to transmit Accelerometer, Barometer, and GPS telemetry via a LoRa channel.

Portable DJ Set | *Questa, Lattice Radiant, SEGGER* October 2025 – December 2025
- Implemented real-time dynamic sampling audio processing by interfacing an MCU with an FPGA for DSP.
- Designed a joystick controller for playback speed adjustment and FIR filtering levels.